impl PrivacyProfile::to_tradeoff

Michael Shoemate

April 30, 2026

This proof resides in **"contrib"** because it has not completed the vetting process.

1 Hoare Triple

Precondition

Compiler-verified

- Receiver `self` of type `PrivacyProfile`

Caller-verified

The receiver is a valid privacy profile.

Pseudocode

```
def privacyprofile_to_tradeoff(profile: PrivacyProfile) -> TradeoffCurve:
    if profile.epsilon_grid is not None:  #
        eps_samples = list(profile.epsilon_grid)
    else:
        eps_hi = EPS_MAX_START
        while profile.delta(eps_hi) > 1e-12 and eps_hi < 1e6:
            eps_hi *= 2.0
        eps_samples = [eps_hi * i / EPS_GRID_POINTS for i in range(EPS_GRID_POINTS + 1)]  #

    supports = []
    last_delta = None

    for eps in eps_samples:  #
        delta = profile.delta(eps)
        delta_exact = RBig.try_from(delta)

        # If delta is unchanged and epsilon increased, the new support is pointwise worse.
        if last_delta == delta_exact:  #
            continue

        tradeoff = approximate_to_tradeoff(eps, delta)
        supports.append(tradeoff)  #
        last_delta = delta_exact

    def tradeoff_curve(alpha: float) -> float:
        if not (is_finite(alpha) and 0.0 <= alpha <= 1.0):
            raise ValueError("alpha ({alpha}) must be a finite number in [0, 1]")

        if alpha == 0.0:  #
```

```
30            return 1.0
31        if alpha == 1.0:  #
32            return 0.0
33
34        alpha = RBig.try_from(alpha)
35        best = rbig(0)
36
37        for support in supports:  #
38            best = best.max(support(alpha))
39
40        result = f64_neg_inf_cast(best)
41        return result.clamp(0.0, 1.0)  #
42
43    return TradeoffCurve(tradeoff_curve)
```

Postcondition

Theorem 1.1. The pseudocode returns a valid tradeoff curve $\beta(\alpha) = \max_i \beta_i(\alpha)$ where each β_i is a conservative lower bound on the f-DP tradeoff induced by the privacy profile.

Proof. Let $\mathcal{E} = \{\epsilon_1, \epsilon_2, \ldots, \epsilon_n\}$ be the set of sampled epsilon values (determined by lines 2–8). The choice of grid doesn't matter for privacy. Let $\delta_i = \delta(\epsilon_i)$ be the corresponding delta values from the profile.

For each distinct δ_i encountered in the grid (loop starting at line 13, support appended at line 22), the algorithm constructs $\beta_i =$ `approximate_to_tradeoff`(ϵ_i, δ_i) and appends it to the list `supports`. By the proof of `approximate_to_tradeoff`, each β_i is a conservative lower bound on the f-DP tradeoff curve induced by the approximate-DP guarantee (ϵ_i, δ_i).

Line 18 skips any samples where $\delta(\epsilon) = \delta(\epsilon')$ for some $\epsilon < \epsilon'$ already processed. This is sound because for fixed δ, increasing ϵ strictly weakens the approximate-DP constraint: the two affine branches of the approximate-DP tradeoff curve (described in `approximate_to_tradeoff`) are both pointwise smaller when ϵ increases.

The returned tradeoff curve evaluates to $\beta(\alpha) = \max_i \beta_i(\alpha)$ for $\alpha \in (0, 1)$ (loop at line 37), with boundary values $\beta(0) = 1$ (line 29) and $\beta(1) = 0$ (line 31). The pointwise maximum of conservative lower bounds is again a conservative lower bound, so the result is a valid conservative tradeoff curve.

Line 41 clamps the result to $[0, 1]$, which preserves conservatism since we only decrease larger-than-1 values and increase smaller-than-0 values, both of which are measurement errors rather than true losses in conservatism. $\square$